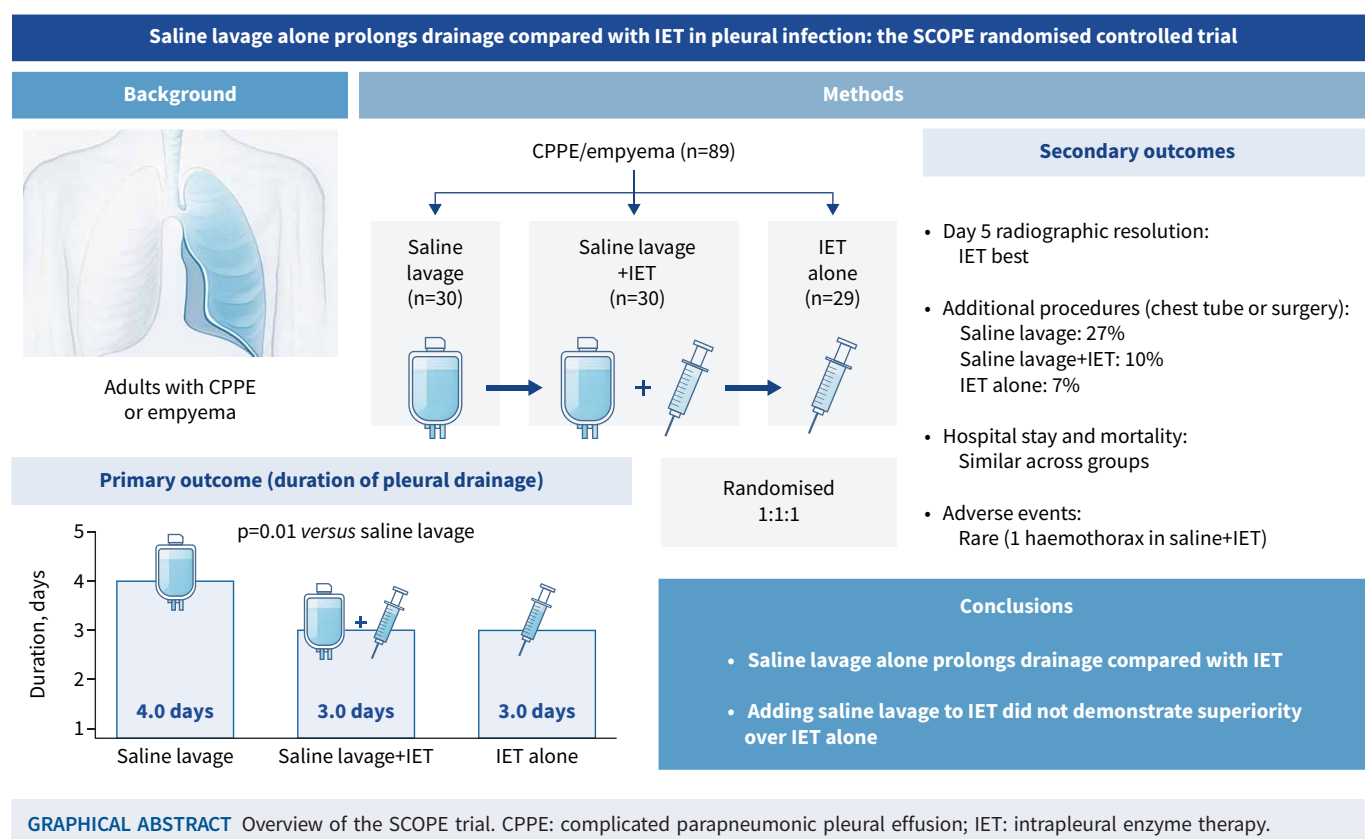


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Saline lavage alone prolongs drainage compared with intrapleural enzyme therapy in pleural infection: the SCOPE randomised controlled trial

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Shareable abstract (@ERSpublications)

In pleural infections, saline lavage alone is associated with a longer drainage duration than intrapleural enzyme therapy (IET). In this small randomised superiority trial, adding saline lavage to IET conferred no meaningful advantage over IET alone. <https://bit.ly/4jXdIBl>

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Abstract

Background Intrapleural enzyme therapy (IET) is widely used for pleural infections, including complicated parapneumonic effusions and empyema; however, the role of saline lavage alone or in combination with IET remains uncertain.

Methods The SCOPE trial was a two-centre, prospective, randomised superiority study designed to determine whether saline lavage alone or combined with IET was superior to IET alone in adults with pleural infection. Patients were randomised in a 1:1:1 ratio to saline lavage alone, saline lavage plus IET (urokinase and DNase) or IET alone. Patients and outcome assessors were blinded to the treatment allocation. The primary end-point was the duration of pleural drainage. Secondary end-points included radiographic resolution, need for additional interventions, length of hospital stay, mortality and adverse events.

Results 89 patients were analysed in three groups (saline lavage, n=30; saline+IET, n=30; IET, n=29). Baseline characteristics were broadly similar across the groups. The median (interquartile range) drainage duration was longer with saline lavage alone (4.0 (3.0–6.75) days) than with IET alone (3.0 (2.0–4.0) days; p=0.01) or saline lavage plus IET (3.0 (3.0–3.75) days; p=0.01). No significant difference was observed between the two IET-containing regimens (p=0.24). Secondary outcomes showed no clear advantage for saline lavage alone or in combination with IET.

Conclusions In pleural infection, saline lavage alone results in a longer drainage duration than IET. In this small superiority trial, adding saline lavage to IET did not demonstrate a clinically meaningful advantage over IET alone.

Introduction

Pleural space infection predominantly arises from underlying pneumonia, with pleural effusion detectable by ultrasonography in approximately half of affected patients [1]. Nevertheless, only a small proportion of parapneumonic pleural effusions (up to 30%) are classified as complicated (CPPEs) [2], indicating that they require drainage, which is commonly performed using a chest tube.

The management of CPPE varies between centres. Several randomised controlled trials (RCTs) have advocated the use of intrapleural enzyme therapy (IET) as a minimally invasive treatment to avoid surgical procedures such as video-assisted thoracoscopic surgery (VATS) [3, 4]. IET consists of a combination of fibrinolytic agents (*e.g.* urokinase or alteplase) and DNase. Alternative therapeutic methods, such as



intrapleural saline administration, have not been sufficiently explored. Only one small RCT has evaluated the potential usefulness of intrapleural saline irrigation as a conservative measure for managing CPPE [5]. Although some benefits were found, the comparator arm exclusively used a chest tube, which is no longer an accepted practice for treating pleural infections.

To date, no study has investigated whether intrapleural saline alone or in combination with IET confers any advantage over the isolated use of IET, which is considered the standard of care for CPPE in numerous medical centres. This constitutes the objective of the present study.

Methods

Study design

The SCOPE (Saline or COmbination intrapleural therapy for Parapneumonic Effusions) trial was a two-centre, prospective, randomised superiority trial with a 1:1:1 allocation ratio. The trial was conducted in adults with pleural infection at two Spanish hospitals (Hospital Universitari Arnau de Vilanova, Lleida and Hospital Universitario Santiago de Compostela, Santiago de Compostela). For the purpose of this study, pleural infection was defined as the presence of pleural fluid consistent with CPPE or empyema, irrespective of radiographic pneumonia.

The trial was designed to determine whether saline lavage, administered alone or in combination with IET, was superior to IET alone, which was considered standard care. The prespecified primary comparison was saline lavage plus IET *versus* IET alone, with the saline-only arm included as an exploratory comparator. This design was informed by prior retrospective observations suggesting that the addition of saline lavage to fibrinolytic therapy could shorten chest tube duration in pleural infection [6].

Randomisation was performed using a computer-generated block design stratified by centre. Patients, treating clinicians and investigators assessing clinical outcomes were blinded to treatment allocation. The trial was registered with the EU Clinical Trials Register (EudraCT) (EudraCT: 2019-003542-32) and received ethical approval from both participating centres (CEIC 2182). All participants provided written informed consent.

Participants

Patients referred to Pleural Medicine Services for thoracentesis because of suspected pleural infection between September 2020 and March 2023 were screened for eligibility. Adults aged ≥ 18 years were included if pleural fluid analysis was consistent with CPPE or empyema, defined by at least one of the following: macroscopically purulent pleural fluid, pleural fluid pH < 7.2 or glucose $< 60 \text{ mg} \cdot \text{dL}^{-1}$, or a positive pleural fluid culture. Radiographic (pleural effusion occupying $\geq 50\%$ of the hemithorax) and ultrasonographic (loculated or septated effusion) features were used as indicators of disease severity and complexity, but were not considered diagnostic criteria in the absence of supportive pleural fluid biochemical or microbiological findings.

The exclusion criteria were pregnancy, allergy to urokinase or DNase, survival expectancy of < 3 months, recent major surgery or trauma, active bleeding, stroke within the past 3 months, tuberculous pleuritis, or pleural infection secondary to recent thoracic surgery or trauma.

Interventions

The day of chest tube insertion and randomisation was considered as trial day 0. Small-bore chest tubes ($\leq 14\text{F}$), inserted using the Seldinger technique and under ultrasound guidance, were primarily used. Following chest tube insertion, fluid was aspirated to the maximum volume that the patient could tolerate without experiencing chest tightness or persistent coughing, or until fluid outflow ceased. Aspiration was performed manually or using a Redon vacuum bottle drainage system. Subsequently, the patients were randomly assigned to one of the three treatment arms. Patients allocated to the saline lavage group underwent treatment involving manual instillation of 0.9% saline through the chest tube, followed by immediate aspiration (either manually or *via* vacuum bottle drainage) until the fluid was clear. This procedure consisted of repeated manual instillation–aspiration cycles performed during the same treatment session rather than a single fixed-volume instillation. For patients randomised to receive the combination of saline and IET (saline lavage+IET group), following saline instillation as previously described, 100 000 U urokinase and 10 mg DNase were administered concurrently through the chest catheter, with a subsequent dwell time of 1.5 h before drainage. This drainage was conducted manually or with a vacuum bottle if the patient had a chest tube inserted in the posterior thorax, or by allowing the fluid to drain into a three-compartment drainage system for those patients whose tube thoracostomy was positioned in the

lateral or anterior thorax. Finally, a third group of patients received only IET as previously described (IET group). The procedures within each treatment arm were repeated once daily for a maximum of 6 days.

The decision to remove the chest tube (which defined the primary outcome of drainage duration) was made daily by a senior pleural specialist at each centre who was blinded to the treatment allocation. To standardise this process across sites, predefined criteria were applied: 1) ultrasound evidence of minimal residual effusion without re-accumulation; 2) clinical stability, defined by improvement in respiratory symptoms and resolution of fever; and 3) a favourable biomarker trajectory, specifically a declining peripheral leukocyte count and serum C-reactive protein level. All three criteria had to be fulfilled before chest tube removal, which occurred on the first day when all criteria were simultaneously met. Persistent low-volume drainage was not used as a criterion for tube retention, as it may continue despite adequate radiological resolution and does not necessarily reflect ongoing infection.

All patients with CPPE received broad-spectrum empiric antibiotic treatment, mostly amoxicillin-clavulanate [7]. Antibiotics were initially administered intravenously and switched to oral administration when the patient became afebrile. The minimum duration of antibiotic therapy was 2 weeks, which could be extended by 1 or 2 weeks depending on the clinical, radiological or analytical circumstances [8].

Outcomes

The primary outcome was the number of days required for pleural drainage, defined as the time from randomisation to chest tube removal. Secondary outcomes included reduction in pleural effusion size on chest radiography on days 2 and 5 (with day 0 defined as the day of randomisation), need for additional chest tube placement or surgery, length of hospital stay, mortality and adverse events within 90 days.

Data collection

Baseline clinical data were collected at enrolment and included demographics (age and sex), smoking habits, comorbidities, use of anticoagulant/antiplatelet therapies, vital signs, symptoms, RAPID (Renal function, Age, Purulence, Infection source and Dietary factors) score [9, 10], percentage of the hemithorax occupied by the pleural effusion on chest radiography, echogenicity pattern of pleural fluid, and macroscopic appearance and analyses of pleural fluid. The area of pleural opacity on chest radiographs was measured digitally using a Java image-processing programme developed by the National Institutes of Health (ImageJ; <https://imagej.net/ij>). Follow-up data were collected on days 2, 5, 30 and 90, and referred to symptoms, microbiological results of the pleural fluid, and primary and secondary outcomes of interest.

Statistical analysis

The sample size was calculated for a superiority design, with the primary comparison being saline lavage plus IET *versus* IET alone. Based on our previous retrospective study [6], mean $\pm$ SD drainage duration was 6.6 $\pm$ 5.6 days with IET alone and 2.3 $\pm$ 3.1 days with saline lavage plus IET, corresponding to an expected difference of $\sim$ 4 days. These estimates were obtained from a reanalysis of the original patient-level data rather than relying on published summary statistics. Assuming a common standard deviation of 4.8 days, 80% power and a two-sided α of 0.05, 27 patients were required per group. Given the three-arm (1:1:1) design, this sample size was applied to all treatment arms (81 patients in total). The calculation was based on detecting a difference in the total duration of pleural drainage rather than differences at a prespecified follow-up time-point. Accordingly, the study was not powered to formally assess non-inferiority.

Quantitative variables were summarised using median (interquartile range (IQR)) and qualitative variables were described using frequency (percentage). The Kruskal–Wallis test was applied for three-group comparisons. When significant differences were identified, pairwise comparisons were performed using the Mann–Whitney U-test with Benjamini–Hochberg correction. For qualitative variables, the Chi-squared test was used, or when the expected frequencies were <5 , Fisher's exact test. Quantile regression was used to model the median post-treatment pleural effusion percentage on days 2 and 5, including treatment group, baseline effusion size and their interaction to assess treatment effects over time. Standard errors were estimated using bootstrap resampling with 200 replicates. The time trend (days 2 and 5) was evaluated by including the visit time as an interaction term in the same regression model. Statistical analyses were performed using R software (www.r-project.org), with a significance level of $p < 0.05$. All analyses were conducted according to the intention-to-treat principle and a prespecified Statistical Analysis Plan (SAP). The SAP, aligned with the EudraCT protocol (2019-003542-32), is provided in the supplementary material.

Results

Baseline characteristics

A total of 252 patients were screened for their eligibility. 90 patients were randomised, but one patient was excluded after randomisation because of a diagnosis of tuberculous pleuritis. Thus, 89 patients were included in the final analysis in the three groups: saline lavage (n=30), saline lavage+IET (n=30) and IET (n=29) (figure 1). The median age of the cohort was 62 years; 73% were male, nearly 80% of infections were community-acquired and approximately one-third corresponded to empyema (macroscopic pus). In >90% of cases, a small-bore chest tube was placed, and the median duration of antibiotic treatment was ~3 weeks. The volume of pleural fluid drained during treatment was ~1 L across groups, whereas the cumulative volume of intrapleural saline administered was 1800 mL in the applicable arms. Baseline demographic and clinical characteristics are summarised in table 1.

Primary outcome

The distribution of pleural drainage duration differed among the three treatment groups (Kruskal–Wallis test, $p=0.0058$). The median (IQR) number of days with pleural drainage was significantly longer in the saline lavage group (4.0 (3.0–6.75) days) than in the IET (3.0 (2.0–4.0) days; $p=0.01$) and saline lavage+IET (3.0 (3.0–3.75) days; $p=0.01$) groups. No difference was observed between the combination and IET groups ($p=0.24$), indicating that the addition of saline lavage did not result in a shorter drainage duration compared with IET alone (figure 2).

Secondary outcomes

By day 2, all three treatments had reduced pleural effusion size to a similar extent, with a median reduction of ~30% from baseline ($p=0.49$). By day 5, IET alone was associated with a greater reduction in effusion size than saline lavage or combination therapy, with a median residual effusion of 0, even for the largest baseline pleural effusions, as determined by quantile regression analysis (interaction $p=0.0026$) (figure 3).

There was a non-significant trend toward a lower requirement for new chest tubes or surgery in the IET group (table 2). Overall, only two patients (2.2%) in the study, both in the combination group, required VATS debridement. Median hospital stay was longer in the saline lavage group than in the other groups

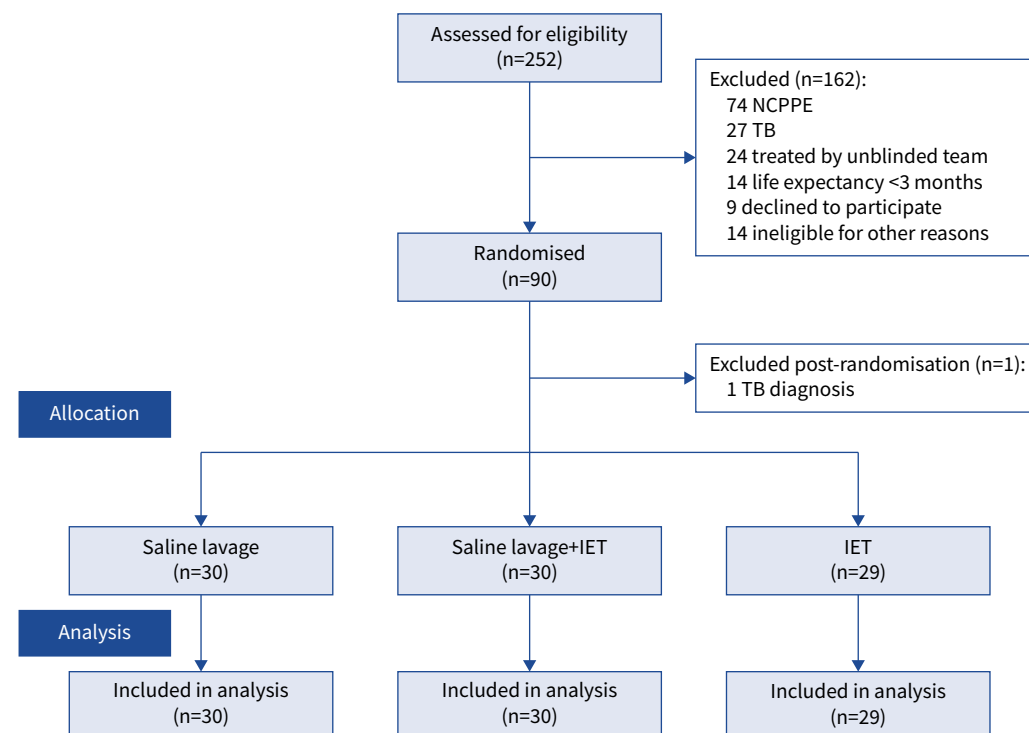


FIGURE 1 CONSORT (Consolidated Standards of Reporting Trials) flow diagram of patient screening, randomisation and analysis in the SCOPE trial. NCPPE: non-complicated parapneumonic effusion; TB: tuberculosis; IET: intrapleural enzyme therapy.

TABLE 1 Patient characteristics by treatment group

	Saline lavage (n=30)	Saline lavage+IET (n=30)	IET (n=29)
Age, years	62 (51–74)	62 (50–76)	61 (58–75)
Female	5 (17)	10 (33)	9 (31)
Current smoker	14 (47)	7 (23.3)	11 (37.9)
Comorbidities			
COPD	4 (13.3)	4 (13.3)	5 (17.2)
Heart failure	4 (13.3)	4 (13.3)	2 (6.9)
Chronic kidney disease	2 (6.7)	2 (6.7)	1 (3.4)
Diabetes	11 (37)	8 (27)	5 (17)
Use of antiplatelet drugs	5 (17)	2 (7)	7 (24)
Use of anticoagulant therapy	4 (13)	5 (17)	4 (14)
Community-acquired pleural infection	21 (70)	24 (80)	25 (86)
RAPID score			
0–2 (low)	10 (33)	13 (43)	10 (34.5)
3–4 (moderate)	9 (30)	7 (23)	14 (48)
5–7 (high)	11 (37)	10 (33)	5 (17)
Pleural effusion size, % on chest radiography	45.5 (29–58)	36.5 (29–59)	37 (29–43)
Pleural fluid pattern on ultrasound			
Complex non-septated	1 (3)	1 (3)	2 (7)
Complex septated	18 (60)	22 (73)	19 (66)
Homogeneously echogenic	11 (37)	7 (23)	8 (28)
Pleural fluid characteristics			
Pus	10 (33)	9 (30)	9 (31)
pH	6.93 (6.80–7.14)	6.91 (6.80–7.34)	6.96 (6.85–7.26)
Glucose, mg·dL ⁻¹	10.5 (10–73.8)	32.5 (10–81.8)	59 (28.2–95)
C-reactive protein, mg·L ⁻¹	102 (55–145)	100 (52–153)	111 (78–160)
Positive culture	15 (50)	16 (53)	15 (52)
Chest tube size			
≤14F	28 (93)	29 (97)	24 (83)
>14F	2 (7)	1 (3)	5 (17)
Duration of antibiotic therapy, days	19 (15–23)	20 (15–29)	21 (21–30)
Volume of pleural fluid drained, mL	1008 (453–2010)	940 (648–1380)	1050 (590–1415)
Volume of saline administered, mL	1800 (1200–3075)	1800 (1200–2100)	NA
Doses of IET, n	NA	3 (3–3.75)	3 (2–4)

Data are presented as median (interquartile range) or n (%). IET: intrapleural enzyme therapy; RAPID: Renal function, Age, Purulence, Infection source and Dietary factors [9]; NA: not applicable.

(8 versus 6 days), although this difference was not statistically significant ($p=0.29$). The 3-month mortality was low and similar across groups ($p=0.12$). Major adverse effects were rare; one case of haemothorax occurred in the combination therapy group and required VATS intervention.

Discussion

In this two-centre RCT, we compared three intrapleural strategies for the management of pleural infection: saline lavage alone, saline lavage combined with IET and IET alone. Designed as a superiority study, SCOPE showed that saline lavage alone was associated with a longer duration of pleural drainage than either IET-containing regimen. In contrast, the addition of saline lavage to IET did not demonstrate a clinically meaningful reduction in drainage duration compared with IET alone. These findings reinforce IET as the preferred intrapleural therapy for pleural infection and do not provide evidence to justify the routine addition of saline lavage or its use as a replacement for IET.

Our results contrast with those of the only previous RCT evaluating intrapleural saline irrigation, the Pleural Irrigation Trial (PIT) [5]. In that single-centre pilot RCT, 35 patients with pleural infection were randomised to standard care or saline irrigation with 250 mL three times daily for 3 days. Saline irrigation reduced the computed tomography-measured pleural fluid volume on day 3 and was associated with a lower rate of surgical referral (11% versus 47%). However, the comparator arm received chest tube drainage alone without IET, an approach that is no longer considered optimal, given the robust evidence supporting fibrinolytic–DNase combinations [11]. Moreover, the unusually high surgical referral rate in the PIT control group raises concerns regarding generalisability. In contrast, all patients in SCOPE received an

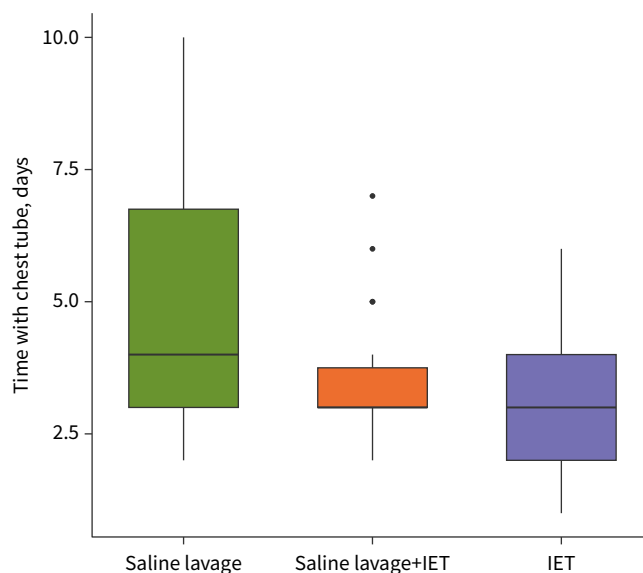


FIGURE 2 Duration of chest tube drainage by treatment group. Box-and-whisker plots show median and interquartile range (IQR); whiskers represent values within 1.5×IQR. IET: intrapleural enzyme therapy.

active, contemporary comparator, allowing a clinically relevant assessment of whether saline offers incremental advantages beyond IET-based care.

Several retrospective studies have suggested a possible benefit of intrapleural saline administration. GUINDE *et al.* [12] reported outcomes in 30 patients with severe empyema, many with advanced cancer, treated with daily pleural lavage *via* small-bore chest tubes for a median of 14 days. Treatment failure at 3 months was relatively low (13%) and hospitalisation was prolonged (median 17 days); however, the absence of a control group limits the inference. More recently, LIU *et al.* [13] retrospectively evaluated 76 patients with

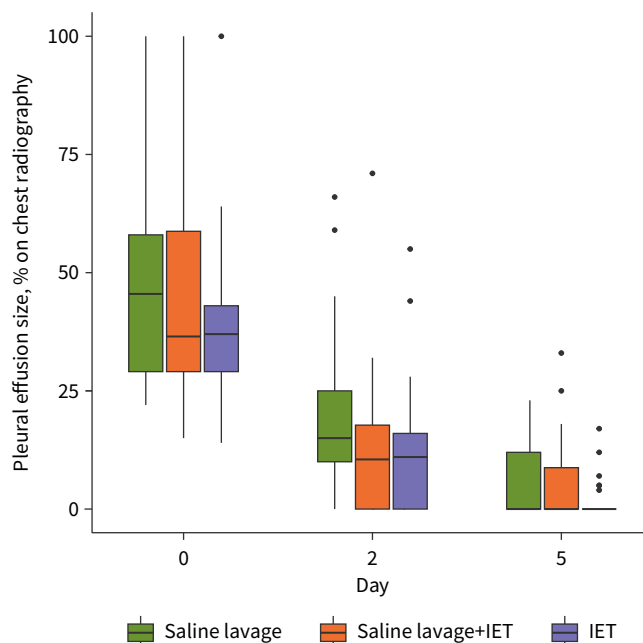


FIGURE 3 Percentage of pleural effusion on chest radiography at baseline (day 0), day 2 and day 5 (with day 0 defined as the day of randomisation) by treatment group. Box-and-whisker plots show median and interquartile range (IQR); whiskers represent values within 1.5×IQR. IET: intrapleural enzyme therapy.

TABLE 2 Secondary outcomes of the SCOPE trial

	Saline lavage	Saline lavage+IET	IET	p-value
Need for additional chest tubes	8 (26.7)	3 (10)	2 (6.9)	0.12
Need for surgery [#]	0 (0)	2 (6.7)	0 (0)	0.32
Length of hospital stay, days	8 (5–12)	6 (4–9)	6 (5–7)	0.29
Infection-related mortality at 3 months [¶]	1 (3)	1 (3.3)	0 (0)	1.0
Adverse events ⁺	1 (3)	1 (3)	1 (3.5)	1.0

Data are presented as n (%) or median (interquartile range), unless otherwise stated. IET: intrapleural enzyme therapy. [#]: all patients who underwent surgery previously required additional chest tubes; [¶]: during follow-up, there were five additional deaths unrelated to pleural infection in the saline lavage group (COVID-19, ischaemic colitis, severe hypoglycaemia, oesophageal cancer and contralateral pneumonia 2 months after resolution of pleural infection), one in the combined saline lavage+IET group (COVID-19) and one in the IET group (aspiration pneumonia) (p=0.136); ⁺: two cases of subcutaneous emphysema (saline lavage and IET groups) and one case of haemothorax (saline lavage+IET group).

parapneumonic empyema treated without IET, comparing conventional drainage with more complex saline-based strategies, including a continuous lavage system using large-bore chest tubes. Although the active lavage system shortened drainage duration compared with simpler irrigation approaches, the mean time with a chest tube remained prolonged (~18 days), substantially longer than that observed in contemporary IET-based regimens, including SCOPE.

In a prior retrospective study from our group [6], 23 patients treated with manual saline flushing plus urokinase were compared with 39 patients receiving urokinase alone. The combined approach was associated with fewer fibrinolytic doses, shorter drainage duration and reduced hospital stay. These observations led to the hypothesis that saline might enhance fibrinolytic efficacy by clearing debris before enzyme instillation. However, this benefit was not reproduced in SCOPE, likely because drainage was already highly optimised in both IET-containing arms through ultrasound-guided tube placement, complete initial aspiration and daily reassessment, leaving little room for additional mechanical clearance.

Beyond retrospective data, recent prospective observational evidence supports the limited efficacy of saline-based strategies. KAWATAKI *et al.* [14] prospectively evaluated 33 patients with culture-positive parapneumonic empyema treated with either daily large-volume saline lavage (500 mL) or intrapleural urokinase monotherapy without DNase. Although differences in surgical referral were not significant, saline lavage was associated with a longer drainage duration (median 10 *versus* 8.5 days) and a significantly prolonged hospital stay after competing risk analysis. These findings reinforce the results of SCOPE, indicating that saline-based approaches do not achieve the same degree of clinical effectiveness as IET.

The physiological rationale for intrapleural saline use is based on the mechanical clearance of infected material. Accordingly, current European Respiratory Society/European Society of Thoracic Surgeons guidance acknowledges that saline administration may have a role when IET is contraindicated or unavailable, but recommends individualised decision making pending stronger evidence [4]. Guidelines emphasise early effective drainage combined with antibiotics, with escalation to IET or surgery when necessary [4, 15]. In this context, the present data do not support the routine addition of saline to IET.

This study has some limitations. It was conducted in two centres within a single country and included a relatively modest sample size, which limited the statistical power for secondary outcomes and for detecting smaller differences between IET-containing regimens. Accordingly, negative findings should be interpreted cautiously. Although chest tube removal involved clinical judgement, variability was minimised through predefined criteria and blinded adjudication by senior pleural specialists. We used urokinase rather than tissue plasminogen activator (tPA) for fibrinolysis, reflecting common practice in many centres [16], but differing from the MIST-2 regimen, which employed sequential tPA–DNase twice daily for 3 days [17]. Our protocol instead used concurrent urokinase and DNase once daily for up to 6 days. Given the substantial heterogeneity in real-world IET practices [18], we cannot exclude the possibility that alternative fibrinolytic–DNase regimens might have yielded different results. Similarly, the saline lavage protocol used in SCOPE (once-daily active instillation with immediate aspiration) differs from those evaluated in prior studies, where saline volumes, frequency and drainage techniques varied considerably [5, 6, 12–14]. Therefore, alternative saline administration strategies may produce different outcomes. Finally, long-term functional outcomes or quality of life were not assessed, and external validation in other populations is warranted.

Conclusions

In this randomised superiority trial with a modest sample size, intrapleural saline lavage was associated with a longer pleural drainage duration than IET alone. The addition of saline lavage to IET did not demonstrate evidence of additional clinical benefit over IET alone when treatment was initiated with optimised drainage under ultrasound guidance and daily reassessment. Despite the limited power for some secondary comparisons, these findings are consistent with IET remaining the preferred intrapleural therapy for pleural infection and do not provide evidence to support the routine use of saline lavage, except in patients with contraindications to fibrinolytic therapy. Further multicentre studies are needed to clarify the role of saline-based strategies, define standardised saline protocols and identify patient subgroups that may benefit.

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Data availability: De-identified individual participant data that underlie the results reported in this article (including text, tables and figures) will be available to researchers upon reasonable request to the corresponding author after publication.

This clinical trial is prospectively registered with the EU Clinical Trials Register (EudraCT) with identifier number 2019-003542-32.

Ethics statement: This study was approved by the local ethics committees of Arnau de Vilanova University Hospital and Santiago de Compostela University Hospital (CEIC 2182) and adhered to the principles outlined in the Declaration of Helsinki.

Author contributions: J.M. Porcel, L. Ferreiro and S. Bielsa conceived and designed this study. J.M. Porcel, L. Ferreiro, C. Acosta, C. Civit and J. Suárez-Antelo were responsible for participant recruitment, clinical evaluation and data collection. L. Rumi was involved in the randomisation process. L. Valdés was involved in the supervision of this study. J.M. Porcel wrote the manuscript. All authors revised and approved the final version of the manuscript.

Conflicts of interest: J.M. Porcel is a Section Editor of the *European Respiratory Journal*. The remaining authors have no potential conflicts of interest to disclose.

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